

The Generation Grading: The Klein Selection Rule, Democratic Gauge Universality, and the Cyclic Generation Symmetry

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Abstract. The covariance audit directed interactions to three representable channels and the charged-coupling paper closed the sector taxonomy with one question left standing: which charges and coupling patterns does the sedenion multiplicity structure permit? This note opens that channel with theorems. The three generation octonions $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3 \subset \mathbb{S}$ have unique directions forming *cosets of the core*: generation i occupies indices $\{i, i+4, i+8, i+12\}$, so the generation label is the index mod 4, and since Cayley-Dickson basis multiplication composes indices by bitwise XOR, the label composes in the Klein four-group $V = \mathbb{Z}_2 \times \mathbb{Z}_2$. **Theorem 1 (the Klein selection rule, verified over all 256 products on an independently rebuilt multiplication table):** $\mathbb{S} = \bigoplus_{v \in V} S_v$ with $S_v S_w \subseteq S_{v \oplus w}$ — same-generation bilinears land in the neutral quaternionic core, and two *distinct* generations couple only into the *third*. **Proposition (democratic gauge universality):** the original gauge-side octonion $\mathbb{O}$ decomposes across the grades as $2+2+2+2$ — it is generation-blind, so any coupling mediated through $\mathbb{O}$ is generation-universal, a graded-decomposition theorem rather than an assumption. **Theorem 2 (the cyclic generation symmetry):** a monomial search over 196,608 candidates (six grade-compatible index maps, all sign assignments) finds *exactly one* automorphism permuting the generations — the pure basis permutation $\tau(e_n) = e_{\sigma(n)}$, all signs $+1$, of order three, fixing the core pointwise while cycling $\mathbb{O}_1 \rightarrow \mathbb{O}_2 \rightarrow \mathbb{O}_3$: the "discrete relativity" of the series' early sedenion papers, now an exhibited $\mathbb{Z}_3 \subset \text{Aut}(\mathbb{S})$. Finally, the kinematic overlap of the generation spaces is *exactly democratic* — the imaginary-part Gram matrix has off-diagonal $3/7$ with eigenvalues $(13/7, 4/7, 4/7)$, S_3 -symmetric with a degenerate pair — so **hierarchical mixing is not kinematic**: the algebra fixes the selection rules and the symmetry, and consigns the numbers, as everywhere in this series, to the state. The interaction program acquires its first octonionic selection rule: Klein-charge conservation for algebra-mediated couplings, universality for gauge-channel couplings, and τ -invariance as a model-building constraint.

1. The channel and its question

The taxonomy of the charged-coupling paper classified interactions by their sector action and ended on the pattern question: which currents carry which charges. The audit's third channel — the sedenion multiplicity structure, the complement's surviving physical employment per the revised opening paper — is where the answer must come from, and this note extracts what the algebra proves.

2. Theorem 1: the Klein grading and the selection rule

Write indices in binary, $n = a + 4b$ with $a \in \{0, 1, 2, 3\}$ the low bits and b the high bits. The generation structure is coset structure: the triple intersection (the core) is $\{0, 4, 8, 12\} = \{a = 0\}$, and generation i 's unique directions are

$\{i, i+4, i+8, i+12\} = \{a = i\}$. The generation label is $\text{gen}(n) = n \bmod 4$, valued in the Klein group $V = \mathbb{Z}_2 \times \mathbb{Z}_2$ with 0 the neutral (core) label.

****Theorem.**** In the Cayley–Dickson basis, $e_m e_n = \pm e_{m \oplus n}$ (bitwise XOR) — verified for all 256 products on a multiplication table rebuilt from the doubling recursion, independently of all prior code — and consequently

$$\mathbb{S} = \bigoplus_{v \in V} S_v, \quad S_v S_w \subseteq S_{v \oplus w}.$$

****Selection rules.**** $S_i S_i \subseteq S_0$: same-generation bilinears are *neutral*, landing in the quaternionic core. $S_i S_j \subseteq S_k$ for $\{i, j, k\} = \{1, 2, 3\}$: two distinct generations couple *only into the third*. S_0 acts within each grade: the core is the generation-preserving channel. ■

This is a conservation law: a Klein-valued generation charge, conserved by the algebra product itself. Any coupling pattern the algebra mediates must respect it.

3. Subalgebras, the core, and democratic gauge universality

Each generation octonion is core plus grade, $\mathbb{O}_i = S_0 \oplus S_i$ — re-verified closed under the full sedenion product on the fresh table — with the core $S_0 = \{1, e_4, e_8, e_{12}\}$ a quaternion algebra, the shared neutral channel of all three. The original octonion $\mathbb{O} = \text{span}(e_0, \dots, e_7)$, by contrast, is *not* grade-aligned: its grade content is $2+2+2+2$ — two directions in every grade, including two in each generation.

Proposition (gauge universality). Any interaction built through $\mathbb{O}$ touches the three generations identically: the gauge-side octonion is generation-blind by its graded decomposition. Universality of gauge-channel couplings is here a theorem of the split, not a phenomenological input. ■

4. Theorem 2: the cyclic generation symmetry, exhibited

The early sedenion papers established the three generations' *relativity* — no canonical one — and this note upgrades relativity to an explicit symmetry. A complete monomial search was run: all index maps σ acting linearly on the bits, inducing a three-cycle on the generation labels and any $GL(2, \mathbb{F}_2)$ transformation of the layer bits, against all 2^{15} sign assignments — 196,608 candidates, each tested against all 256 product relations.

****Theorem.**** Exactly *one* automorphism exists in this family: the pure basis permutation

$$\tau(e_n) = e_{\sigma(n)}, \quad \sigma : \text{low bits } (v_0, v_1) \mapsto (v_1, v_0 \oplus v_1), \text{ layer bits fixed,}$$

with *all signs* $+1$. It has order three, fixes the quaternionic core pointwise, and cycles $\mathbb{O}_1 \rightarrow \mathbb{O}_2 \rightarrow \mathbb{O}_3 \rightarrow \mathbb{O}_1$. ■

A $\mathbb{Z}_3$ generation symmetry inside $\text{Aut}(\mathbb{S})$, with the neutral channel frozen — the cleanest form the discrete relativity could have taken, and its uniqueness within the family adds rigidity: the generation cycle is not one choice among many. Whether a transposition

(completing S_3) also lifts is the same finite search with a swap in place of the cycle, left as a noted extension.

5. Democratic mixing, and the honest boundary

The generation matter spaces $\text{Im } \mathbb{O}_i$ are seven-dimensional with pairwise intersection the three-dimensional Im of the core. Their normalized Gram matrix has diagonal 1 and off-diagonal $3/7$, with eigenvalues $(13/7, 4/7, 4/7)$: the democratic vector and a degenerate doublet, exactly S_3 -symmetric. **Consequence:** any mixing structure derived from the kinematic geometry of the generation subspaces alone is democratic and degenerate — *hierarchy is not kinematic*. The algebra supplies exact structure (the selection rule, the universality, the $\mathbb{Z}_3$); the numbers — masses, mixings, hierarchies — must come from the state stratum and the dynamics, precisely the division of labor this series has found at every level: structure from the algebra, quantities from the state.

6. What the interaction program acquires

Three constraints, now theorems, for model-building within the audit's channels. (i) **Klein-charge conservation:** every algebra-mediated coupling pattern respects $\text{gen}(xy) = \text{gen}(x) \oplus \text{gen}(y)$ — cross-generation transition currents carry the third generation's charge, and the memory-charge bookkeeping of the charged-coupling paper composes accordingly. (ii) **Gauge universality:** couplings through $\mathbb{O}$ are generation-blind by decomposition. (iii) **τ -invariance:** the $\mathbb{Z}_3$ is available as a symmetry to impose or to break — and if broken, broken by the state, not the algebra. The named next construction is the representation of this graded structure on the net: three current families carrying Klein charges, with sources classified by the taxonomy and patterns constrained by the rules above.

7. Scope

Five items. (i) No identification with Standard-Model generations is claimed; the adjacency to the three-generation octonionic literature is carried by the companion papers on Gresnigt's construction and is left there. (ii) The layer bits define a *second* Klein grading (the core's own coset structure), noted as exact structure awaiting employment. (iii) The S_3 completion of the exhibited $\mathbb{Z}_3$ is a finite search, left open. (iv) The automorphism search covered monomial (basis-permutation-with-signs) maps; non-monomial automorphisms are not excluded and not needed for the exhibited symmetry. (v) Hierarchy and mixing values are consigned to the state stratum; nothing here constrains their size — only their symmetry class.

References

- Companion papers: *The Sedenion Split $S = O \oplus O \cdot e_8$: Necessity, Classification, and Discrete Relativity*; *Generation Is Not Color: Resolving the Symmetric Embedding of Algebraic R-Flux in Gresnigt's Three-Generation Construction*; *The Covariance Audit: Why the Native Deformation Cannot Ride the Boosts, and Where Interactions Must Enter*; *The Charged Coupling: The Infrared Catastrophe as a Power Law, Sector Pumping, and the Memory-Charge Relation*.

- Günaydin, M., Lüst, D. & Malek, E., "Non-associativity in non-geometric string and M-theory backgrounds, the algebra of octonions, and missing momentum modes," arXiv:1607.06474.

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